

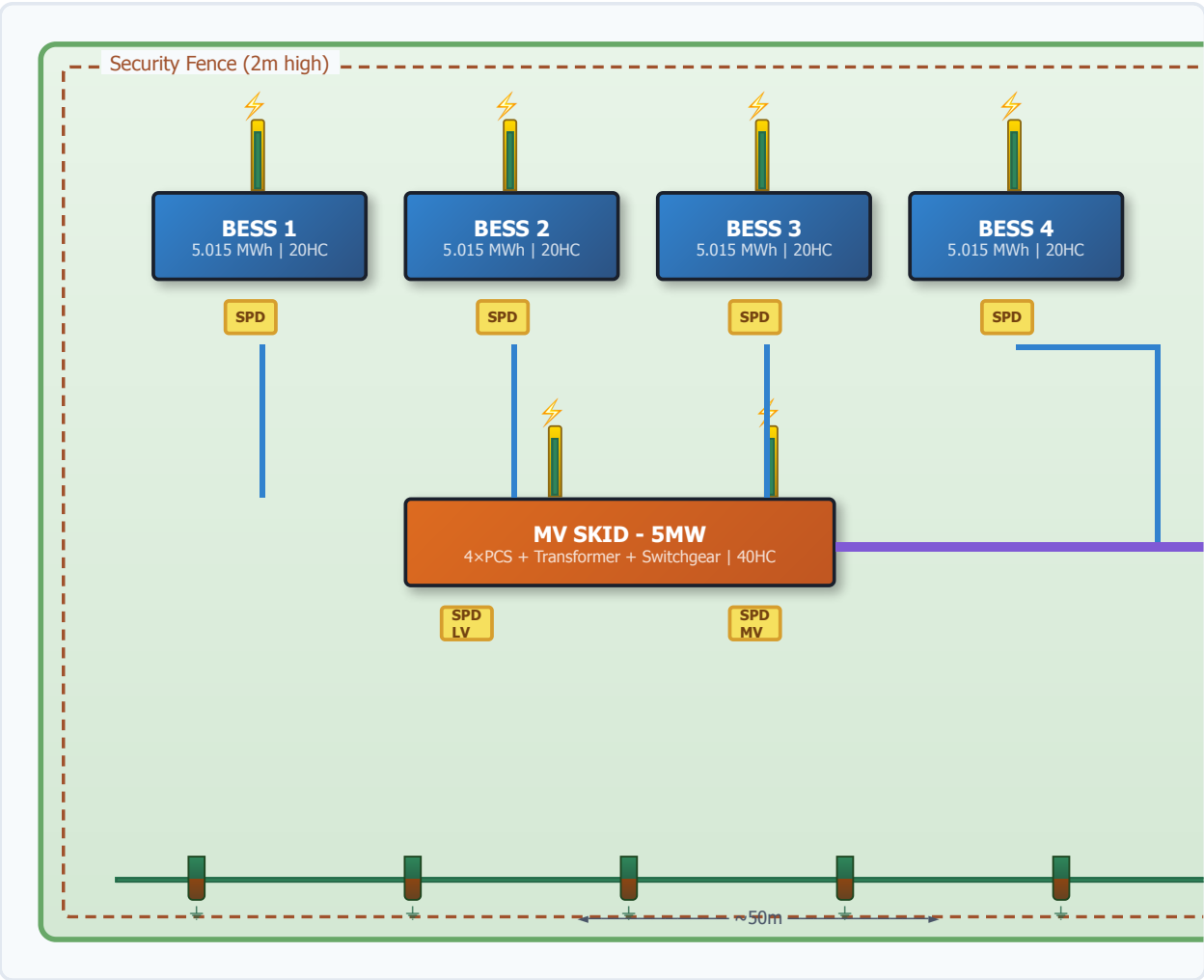
⚡ 5MW/20MWh BESS Park - Lightning Protection Layout

For Dhen - LPS & SPD Design Reference | Lighthief Cyprus Ltd

1. Park Configuration

Parameter	Value	Dimension	Value
Power Rating	5 MW	Container Height	2.896 m
Energy Capacity	20 MWh	20HC (L × W)	6.058 × 2.438 m
Battery Containers	4 × 20HC (5.015 MWh each)	40HC (L × W)	12.192 × 2.438 m
MV SKID	1 × 40HC (5MW)	Site Footprint (est.)	~50 × 30 m
Total Structures	5 units	Perimeter Fence	~160 m

2. Site Layout Diagram (Plan View)



■ Battery Container (20HC) ■ MV SKID (40HC) ■ ⚡ Lightning Rod (Air Terminal)

3. Lightning Protection Requirements

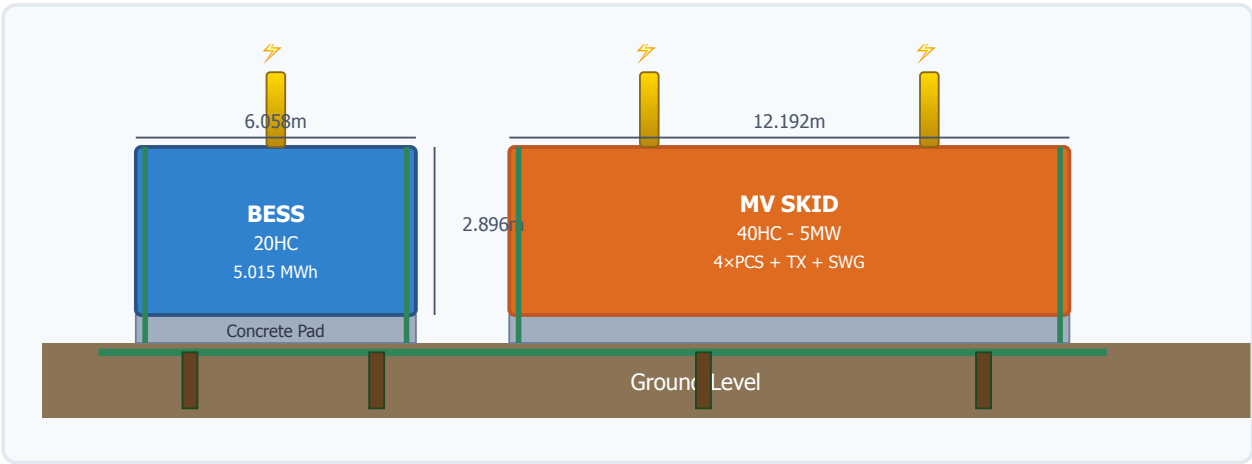
LPS Components Required

Component	Qty	Notes
Air Terminals (Lightning Rods)	6	1 per BESS + 2 for MV SKID
Down Conductors	12	Min 2 per structure
Earth Electrodes	6-10	Per soil resistivity
Earth Grid (Cu tape)	~100m	Ring + cross bonds
Bonding Connections	20+	All metallic structures

SPD Requirements

Location	Type	Voltage
Grid POC (MV incoming)	Type 1	11 kV
MV SKID LV panel	Type 1+2	400 V AC
Each Battery Container	Type 2	1500 V DC
Aux Power (each container)	Type 2	400 V AC
BMS/SCADA/Comms	Type 3	24V DC / Ethernet

4. Container Dimensions (Side View)



5. Electrical Specifications

System	Voltage	SPD Requirement	Location
DC System	1,165 - 1,498 V DC (nominal 1,331 V)	Type 2 SPD, $U_c \geq 1500V$ DC, $I_{imp} \geq 20kA$	Each battery container DC output
LV AC System	400 V AC 3-phase	Type 1+2 SPD, $U_c \geq 440V$ AC, $I_{imp} \geq 12.5kA$	MV SKID LV distribution panel
MV System	11 kV or 22 kV	MV Surge Arresters, $U_r \geq 12kV/24kV$	Grid POC & MV Switchgear
Auxiliary Power	400 V AC 3-phase (4x35mm ² + 1x16mm ²)	Type 2 SPD, $U_c \geq 440V$ AC	Each container aux input
Control/BMS	24 V DC	Type 3 SPD, data line protection	BMS cabinet, SCADA cabinet

Communication	Ethernet CAT6 / STP	Ethernet SPD, PoE compatible	Each Ethernet connection point
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6. Earthing Specifications (Per Linyang)

Parameter	Requirement	Notes
Earth Grid Cross-Section	$\geq 250 \text{ mm}^2$ effective	Cu tape or equivalent
Connection Impedance	$\leq 0.1 \Omega$	Between any two points
Grounding Cable	120-150 mm ² Cu	Yellow-green or flat steel
Target Earth Resistance	$\leq 1 \Omega$	Per container, Cyprus soil conditions
Overall System Earth	$\leq 0.5 \Omega$	Combined park earth resistance

⚠ Note for Dhen: Cyprus soil conditions vary significantly. Rocky/limestone areas may require enhanced earthing (chemical earth, deeper rods, or plate electrodes). Please include contingency for earthing enhancement in your quotation.

7. Quotation Request Summary

Item	Quantity	Unit Price €	Total €
LPS for 20HC Container (complete)	4		
LPS for 40HC MV SKID (complete)	1		
SPD Kit - DC 1500V (per container)	4		
SPD Kit - LV 400V (MV SKID)	1		
SPD Kit - MV 11kV (Grid POC)	1		
SPD Kit - Control/Comms (per container)	5		
Earthing Grid + Electrodes (complete)	1 lot		
Earthing Enhancement (contingency)	1 lot		
Installation Labor	1 lot		
Testing & Certification (BS EN 62305)	1 lot		
TOTAL FOR 5MW/20MWh PARK			

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